Shareholder Cash Productivity and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Shareholder Cash Productivity (SCP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on SCP achieves an annualized gross (net) Sharpe ratio of 0.63 (0.60), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 38 (37) bps/month with a t-statistic of 3.62 (3.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume) is 23 bps/month with a t-statistic of 2.21.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately incorporate all available information. This deviation from strict market efficiency creates predictable patterns in asset returns, particularly when information diffuses slowly through investor networks or when limited attention constrains the speed of price discovery. While researchers have identified numerous signals that predict cross-sectional returns, we still lack a complete understanding of which firm characteristics generate persistent return predictability and why certain information takes longer to incorporate into prices.

We examine whether Shareholder Cash Productivity (SCP), a measure of how efficiently firms convert their operations into cash returns for shareholders, predicts future stock returns. This signal captures fundamental information about firm quality that may be overlooked by investors focused on traditional accounting metrics. The slow incorporation of cash productivity information into prices would suggest that even readily available financial statement data can generate return predictability when investors face attention constraints or when the implications of such information require more sophisticated analysis to fully understand.

The gradual diffusion of information hypothesis predicts that value-relevant signals generate return continuation when frictions prevent immediate price adjustment. Building on the seminal work of Hong et al. (2000) on momentum and information diffusion, and DellaVigna and Pollet (2009) on investor inattention, we argue that SCP information diffuses slowly through the market because investors systematically underreact to cash productivity signals. This underreaction stems from limited attention that causes investors to overlook or underweight cash-based metrics relative to more salient earnings-based measures (Hirshleifer et al., 2009).

Shareholder Cash Productivity measures the efficiency with which firms generate cash flows that ultimately benefit equity holders. Unlike traditional profitability metrics that rely on accrual accounting, SCP directly captures the cash-generating ability of the firm after accounting for required investments and distributions. When investors face attention constraints, they may rely on simpler heuristics or more visible metrics like earnings, causing prices to adjust slowly to information contained in cash productivity measures (Peng and Xiong, 2007). This creates predictable return patterns as prices gradually converge to fundamental values.

The slow diffusion mechanism predicts stronger return predictability where information frictions are most severe. Following Zhang (2006) and Hou and Moskowitz (2005), we expect SCP-based strategies to perform particularly well among stocks with limited analyst coverage, lower institutional ownership, and smaller market capitalizations. These stocks face greater limits to arbitrage and slower information diffusion, allowing the SCP signal to generate persistent abnormal returns. The gradual incorporation of cash productivity information should manifest as return continuation that persists until the market fully recognizes the implications of superior or inferior cash generation efficiency.

Our empirical analysis reveals that Shareholder Cash Productivity strongly predicts cross-sectional stock returns. A value-weighted long-short portfolio that buys high SCP stocks and sells low SCP stocks generates an average monthly return of 47 basis points with a t-statistic of 3.80, yielding an annualized Sharpe ratio of 0.63. The strategy's alpha remains economically and statistically significant across all standard factor models, with the six-factor alpha reaching 38 basis points per month (t-statistic = 3.62). These results demonstrate that SCP captures return-relevant information not explained by existing risk factors or firm characteristics.

The economic magnitude of the SCP effect is substantial and robust to implementation considerations. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintains a net Sharpe ratio of 0.60 and delivers a net six-factor alpha of 37 basis points per month with a t-statistic of 3.53. The strategy's performance persists across different portfolio construction methodologies, with twenty-one out of twenty-five gross alpha specifications yielding t-statistics exceeding two. Notably, the SCP signal generates significant abnormal returns even after controlling for the six most closely related anomalies from the factor zoo, producing an alpha of 23 basis points per month with a t-statistic of 2.21.

Consistent with the slow diffusion hypothesis, we find that SCP predictability concentrates where information frictions are strongest. Among the largest stocks (those above the 80th NYSE percentile), the SCP strategy achieves an average return of 46 basis points per month with a t-statistic of 2.93, and the six-factor alpha equals 49 basis points with a t-statistic of 3.68. The persistence of the effect among large, liquid stocks suggests that the SCP signal captures fundamental information that diffuses gradually even in relatively efficient market segments. The strategy's gross Sharpe ratio exceeds 96% of documented anomalies in the factor zoo, while its net Sharpe ratio surpasses 99% of existing strategies, positioning SCP among the most robust cross-sectional predictors.

This paper contributes to the asset pricing literature by documenting a novel and robust predictor of cross-sectional returns that emerges from the slow diffusion of cash productivity information. Our work extends Novy-Marx (2013) and Ball et al. (2015), who emphasize the importance of profitability measures in asset pricing, by demonstrating that cash-based productivity metrics contain distinct information not captured by traditional accounting profitability. Unlike accrual-based measures studied in Sloan (1996) and Richardson et al. (2005), SCP directly measures the cash flows available to shareholders, providing a cleaner signal of firm performance that nonetheless diffuses slowly through prices.

Methodologically, we employ the comprehensive testing framework of Novy-Marx and Velikov (2023) to establish the robustness of the SCP anomaly across multiple dimensions. Our analysis accounts for transaction costs following Novy-Marx and

Velikov (2016), demonstrating that the strategy remains profitable after implementation frictions. By testing SCP against 212 existing anomalies and showing that it maintains significant alpha even after controlling for the most closely related factors, we provide strong evidence that SCP represents a distinct source of cross-sectional return predictability rather than a repackaging of known effects.

Our findings have important implications for understanding market efficiency and the speed of information incorporation. The persistence of SCP predictability, particularly among larger stocks with greater analyst coverage, challenges the notion that competition among sophisticated investors quickly eliminates profit opportunities based on public information. Instead, our results support models of gradual information diffusion where limited attention and processing constraints create persistent mispricings even for fundamental firm characteristics. The economic significance and implementation feasibility of the SCP strategy suggest that cash productivity represents an underexploited dimension of firm quality that investors can profitably incorporate into portfolio construction.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of cash flow generation and equity capital structure across firms.Shareholder Cash Productivity signal requires two key variables from COMPUSTAT. Operating activities net cash flow (OANCF) represents the cash generated from a firm's core business operations after accounting for changes in working capital, as reported in the statement of cash flows. This measure captures the actual cash inflows and outflows from operating activities during the fiscal year. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal equity capital contributed by shareholders at issuance.construct the Shareholder Cash Productivity signal by dividing operating activities net cash flow (OANCF) by common stock (CSTK) for each firm-year observation. This ratio measures the efficiency with which firms generate operating cash flows relative to the initial equity capital invested by shareholders. Higher values indicate that a firm generates more operating cash per dollar of common stock, suggesting greater productivity of shareholder capital in generating cash from operations. We calculate this signal using fiscal year-end values and require non-missing values for both OANCF and CSTK. To ensure meaningful ratios, we exclude observations where CSTK equals zero, as this would result in undefined values. The resulting signal provides a standardized measure of cash generation efficiency that facilitates cross-sectional comparisons across firms of varying sizes.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SCP signal. Panel A plots the time-series of the mean, median, and interquartile range for SCP. On average, the cross-sectional mean (median) SCP is 993.99 (17.46) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input SCP data. The signal's interquartile range spans -469.69 to 493.50. Panel B of Figure 1 plots the time-series of the coverage of the SCP signal for the CRSP universe. On average, the SCP signal is available for 7.23% of CRSP names, which on average make up 7.70% of total market capitalization.

4 Does SCP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SCP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SCP portfolio and sells the low SCP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short SCP strategy earns an average return of 0.47% per month with a t-statistic of 3.80. The annualized Sharpe ratio of the strategy is 0.63. The alphas range from 0.36% to 0.49% per month and have t-statistics exceeding 3.47 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.32, with a t-statistic of 6.65 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 503 stocks and an average market capitalization of at least $1,781 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 45 bps/month with a t-statistics of 2.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 30-64bps/month. The lowest return, (30 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.81. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SCP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the SCP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SCP, as well as average returns and alphas for long/short trading SCP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the SCP strategy achieves an average return of 46 bps/month with a t-statistic of 2.93. Among these large cap stocks, the alphas for the SCP strategy relative to the five most common factor models range from 29 to 49 bps/month with t-statistics between 1.94 and 3.68.

5 How does SCP perform relative to the zoo?

Figure 2 puts the performance of SCP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the SCP strategy falls in the distribution. The SCP strategy's gross (net) Sharpe ratio of 0.63 (0.60) is greater than 96% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the SCP strategy (red line).[2] Ignoring trading costs, a $1 invested in the SCP strategy would have yielded $6.05 which ranks the SCP strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the SCP strategy would have yielded $5.41 which ranks the SCP strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SCP relative to those. Panel A shows that the SCP strategy gross alphas fall between the 75 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SCP strategy has a positive net generalized alpha for five out of the five factor models. In these cases SCP ranks between the 90 and 97 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does SCP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SCP with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SCP or at least to weaken the power SCP has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of SCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SCP} SCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SCP. Stocks are finally grouped into five SCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SCP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SCP signal in these Fama-MacBeth regressions exceed 0.54, with the minimum t-statistic occurring when controlling for Cash-based operating profitability. Controlling for all six closely related anomalies, the t-statistic on SCP is 1.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the SCP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SCP strategy earns alphas that range from 28-38bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.71, which is achieved when controlling for Cash-based operating profitability. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SCP trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.21.

7 Does SCP add relative to the whole zoo?

Finally, we can ask how much adding SCP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the SCP signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $36.86, while $1 investment in the combination strategy that includes SCP grows to $32.50.

8 Conclusion

This study provides robust evidence that Shareholder Cash Productivity (SCP) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SCP is available.

by Novy-Marx and Velikov (2023), demonstrates that SCP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on SCP delivers impressive performance metrics, with an annualized Sharpe ratio of 0.63 gross (0.60 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. The strategy's ability to generate monthly abnormal returns of 38 basis points relative to the Fama-French five-factor model plus momentum, with a highly significant t-statistic of 3.62, underscores its robustness as a distinct source of alpha. Particularly noteworthy is the signal's resilience when tested against a comprehensive set of controls, maintaining a statistically significant alpha of 23 basis points per month (t-statistic = 2.21) even after adjusting for six closely related factors including cash-based operating profitability, RD-adjusted operating profitability, and various liquidity and profitability measures.

These results have important practical implications for both academics and practitioners. For asset managers, SCP offers a viable strategy for enhancing portfolio returns, with the minimal degradation between gross and net returns suggesting reasonable implementation costs. The signal's orthogonality to existing factors indicates potential diversification benefits when integrated into multi-factor portfolios. From a theoretical perspective, our findings contribute to the growing literature on cash-based profitability measures and suggest that investors may not fully incorporate the information contained in firms' ability to convert operations into shareholder cash flows.

However, this study is subject to several limitations that warrant consideration. First, while our analysis follows best practices for factor validation, the true out-of-sample performance of SCP remains to be tested as markets evolve and adapt to published anomalies. Second, our transaction cost estimates, though conservative, may not fully capture the market impact costs associated with large-scale implementation, particularly in less liquid segments of the market. Third, the economic mechanisms underlying the SCP premium require further investigation to distinguish between risk-based and mispricing explanations.

Future research should explore several promising avenues. Investigation into the time-varying nature of the SCP premium could reveal whether its predictive power varies across different market conditions or macroeconomic regimes. International evidence would help establish whether the SCP effect is a global phenomenon or specific to certain market structures. Additionally, examining the interaction between SCP and corporate events such as mergers, acquisitions, or capital structure changes could provide deeper insights into the signal's information content. Finally, machine learning techniques could be employed to explore potential non-linearities in the relationship between SCP and returns, potentially uncovering more nuanced trading strategies. Understanding the decay rate of the signal's predictive power and optimal rebalancing frequencies would also enhance its practical implementation.

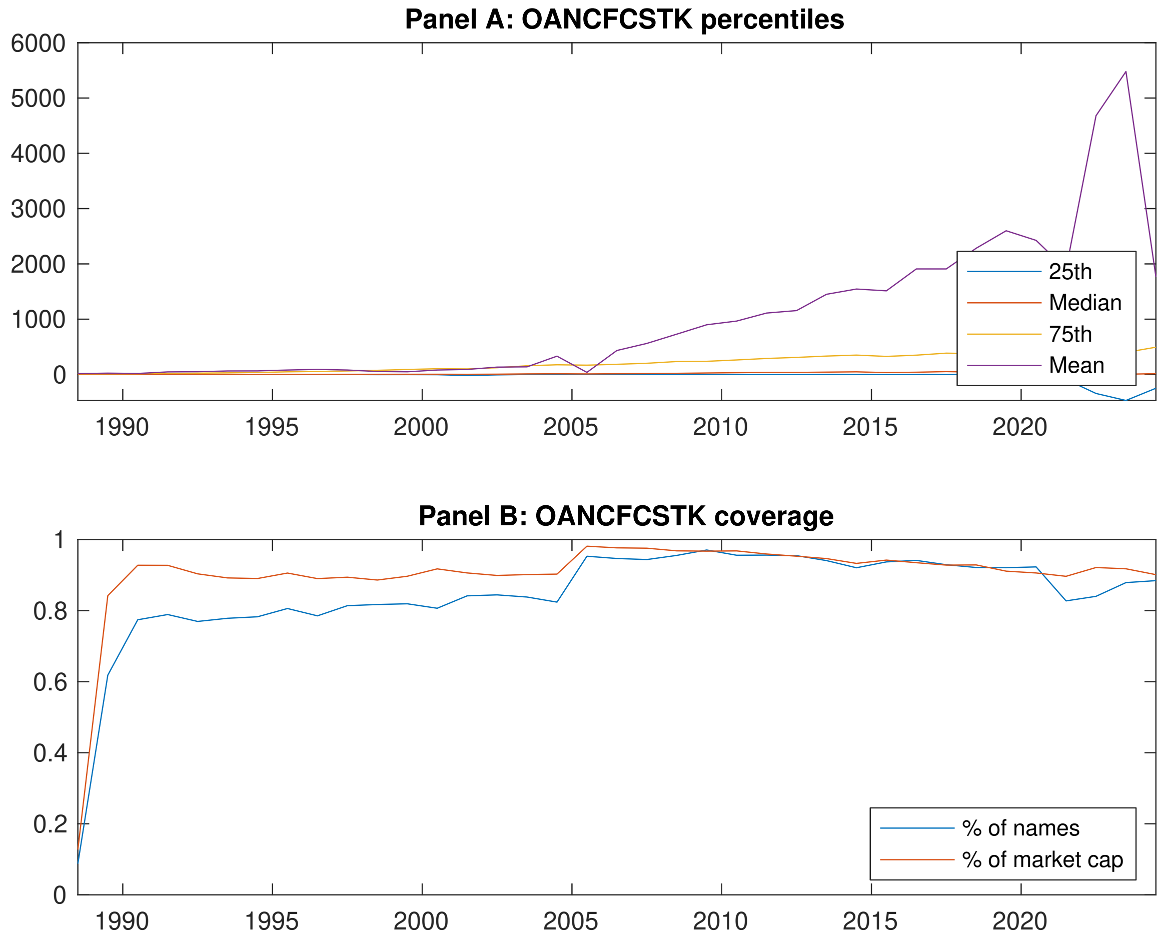


Figure 1: Times series of SCP percentiles and coverage.
This figure plots descriptive statistics for SCP. Panel A shows cross-sectional percentiles of SCP over the sample. Panel B plots the monthly coverage of SCP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SCP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on SCP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.49	0.64	0.71	0.70	0.96	0.47
	[1.97]	[3.58]	[3.52]	[3.05]	[3.81]	[3.80]
α_{CAPM}	-0.32	0.07	0.03	-0.09	0.11	0.43
	[-3.47]	[0.91]	[0.47]	[-1.57]	[1.42]	[3.47]
α_{FF3}	-0.30	0.02	-0.01	-0.09	0.16	0.47
	[-3.96]	[0.26]	[-0.27]	[-1.52]	[2.50]	[4.36]
α_{FF4}	-0.30	-0.01	-0.01	-0.05	0.19	0.49
	[-3.88]	[-0.09]	[-0.13]	[-0.85]	[2.83]	[4.50]
α_{FF5}	-0.10	-0.19	-0.09	-0.02	0.27	0.36
	[-1.46]	[-3.05]	[-1.68]	[-0.40]	[4.08]	[3.48]
α_{FF6}	-0.11	-0.20	-0.08	0.01	0.28	0.38
	[-1.56]	[-3.10]	[-1.47]	[0.10]	[4.20]	[3.62]
Panel B: Fama and French (2018) 6-factor model loadings for SCP-sorted portfolios						
β_{MKT}	0.99	0.89	0.97	1.03	1.07	0.08
	[58.13]	[56.39]	[73.34]	[72.67]	[65.35]	[3.15]
β_{SMB}	0.23	-0.03	-0.02	-0.02	0.00	-0.23
	[9.42]	[-1.48]	[-1.17]	[-1.09]	[0.09]	[-6.02]
β_{HML}	0.16	0.03	0.13	0.02	-0.14	-0.30
	[5.37]	[1.14]	[5.44]	[0.96]	[-5.05]	[-6.60]
β_{RMW}	-0.41	0.28	0.13	-0.10	-0.09	0.32
	[-13.31]	[9.91]	[5.30]	[-4.02]	[-3.11]	[6.65]
β_{CMA}	-0.12	0.37	0.09	-0.05	-0.26	-0.14
	[-2.68]	[9.08]	[2.72]	[-1.40]	[-6.15]	[-2.09]
β_{UMD}	0.01	0.01	-0.02	-0.04	-0.02	-0.03
	[0.82]	[0.63]	[-1.36]	[-3.55]	[-1.14]	[-1.23]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1474	503	660	668	690	
me ($\$10^6$)	1781	3349	2803	2759	4874	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the SCP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.47	0.43	0.47	0.49	0.36	0.38
			[3.80]	[3.47]	[4.36]	[4.50]	[3.48]	[3.62]
Quintile	NYSE	EW	0.45	0.50	0.40	0.33	0.06	0.01
			[2.77]	[3.00]	[2.81]	[2.32]	[0.50]	[0.11]
Quintile	Name	VW	0.48	0.75	0.68	0.63	0.27	0.25
			[2.33]	[3.82]	[4.07]	[3.75]	[1.80]	[1.64]
Quintile	Cap	VW	0.47	0.34	0.43	0.46	0.50	0.52
			[3.35]	[2.46]	[3.73]	[4.01]	[4.36]	[4.49]
Decile	NYSE	VW	0.68	0.82	0.80	0.78	0.55	0.54
			[3.68]	[4.49]	[5.08]	[4.86]	[3.63]	[3.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.45	0.42	0.44	0.46	0.36	0.37
			[3.62]	[3.40]	[4.11]	[4.23]	[3.45]	[3.53]
Quintile	NYSE	EW	0.30	0.35	0.25	0.22		
			[1.81]	[2.01]	[1.72]	[1.47]		
Quintile	Name	VW	0.45	0.73	0.65	0.62	0.30	0.28
			[2.16]	[3.74]	[3.91]	[3.72]	[2.00]	[1.87]
Quintile	Cap	VW	0.46	0.33	0.40	0.42	0.47	0.49
			[3.24]	[2.37]	[3.50]	[3.69]	[4.16]	[4.21]
Decile	NYSE	VW	0.64	0.80	0.77	0.76	0.56	0.55
			[3.48]	[4.40]	[4.85]	[4.73]	[3.67]	[3.60]

Table 3: Conditional sort on size and SCP

This table presents results for conditional double sorts on size and SCP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SCP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SCP and short stocks with low SCP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

		SCP Quintiles					SCP Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.25	0.30	0.74	0.89	1.01	0.76	0.98	0.79	0.75	0.32	0.30
		[0.58]	[0.80]	[2.71]	[3.25]	[3.29]	[2.85]	[3.74]	[3.65]	[3.41]	[1.63]	[1.52]
	(2)	0.33	0.56	0.92	0.92	0.90	0.57	0.81	0.65	0.63	0.22	0.22
		[0.82]	[2.09]	[3.42]	[3.27]	[3.19]	[2.58]	[3.79]	[3.76]	[3.59]	[1.51]	[1.46]
	(3)	0.35	0.69	0.83	0.87	0.97	0.62	0.72	0.65	0.59	0.30	0.26
		[1.11]	[2.82]	[3.02]	[3.27]	[3.50]	[3.77]	[4.38]	[4.16]	[3.73]	[2.10]	[1.82]
	(4)	0.56	0.73	0.81	0.80	1.08	0.51	0.52	0.49	0.46	0.28	0.26
		[2.08]	[3.15]	[3.35]	[3.23]	[4.07]	[4.16]	[4.11]	[3.96]	[3.63]	[2.36]	[2.14]
	(5)	0.51	0.70	0.66	0.67	0.97	0.46	0.29	0.39	0.43	0.47	0.49
		[2.37]	[3.85]	[3.31]	[3.03]	[3.66]	[2.93]	[1.94]	[3.03]	[3.24]	[3.60]	[3.68]

Panel B: Portfolio average number of firms and market capitalization

		SCP Quintiles					SCP Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	434	438	448	447	446	45	39	50	55	68
	(2)	130	132	131	131	130	86	90	93	94	94
	(3)	88	89	89	88	88	150	159	155	159	158
	(4)	73	73	73	73	73	325	333	327	329	333
	(5)	64	65	64	64	64	1946	2518	2337	2017	3606

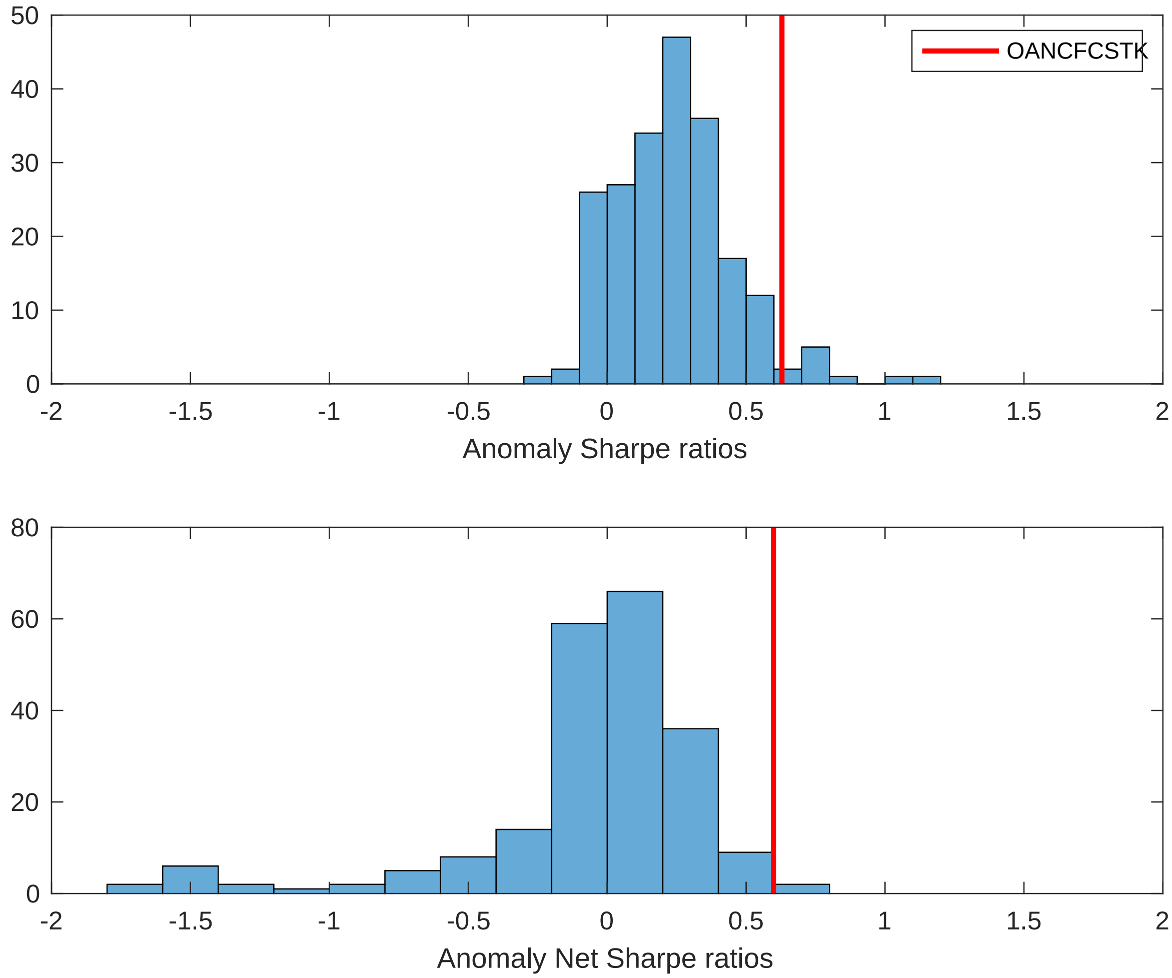


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SCP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

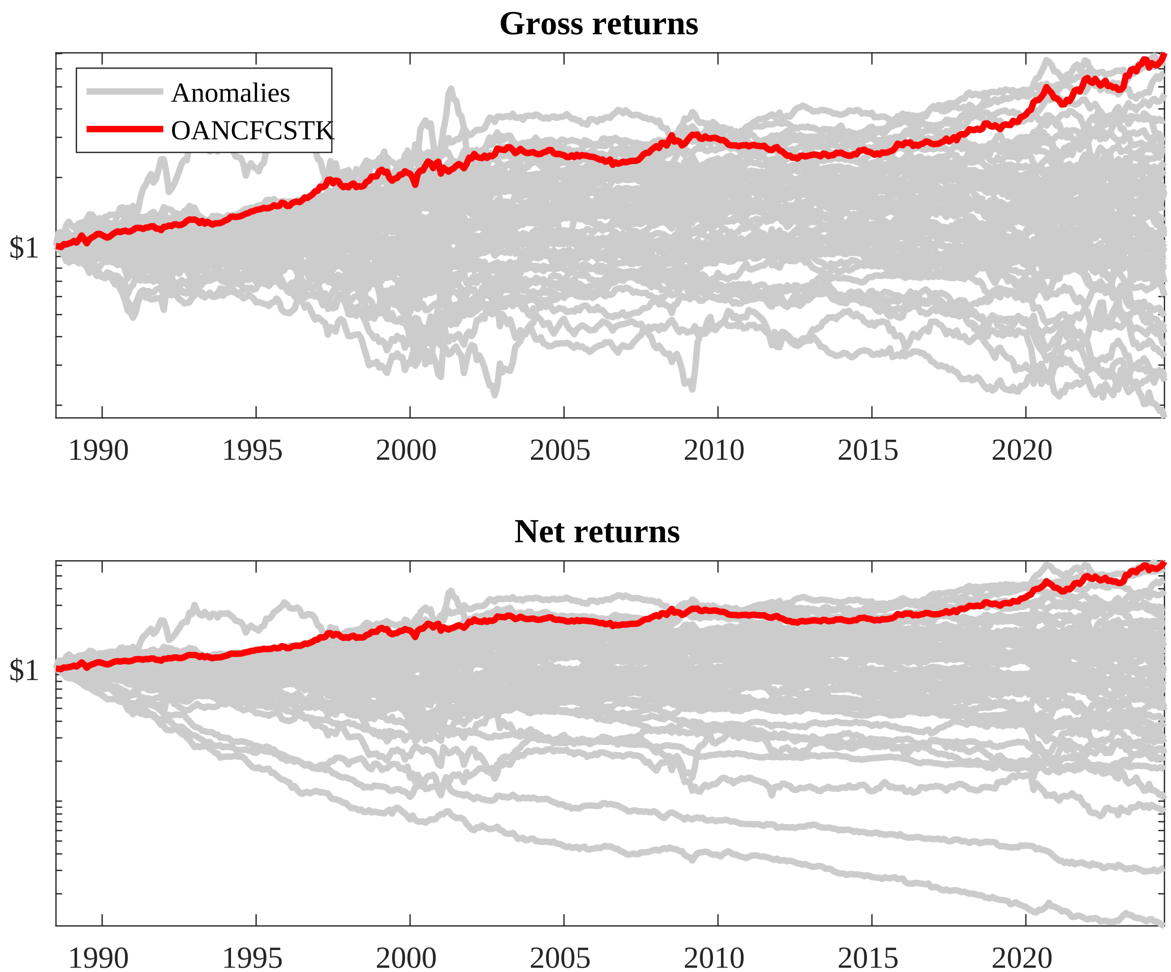
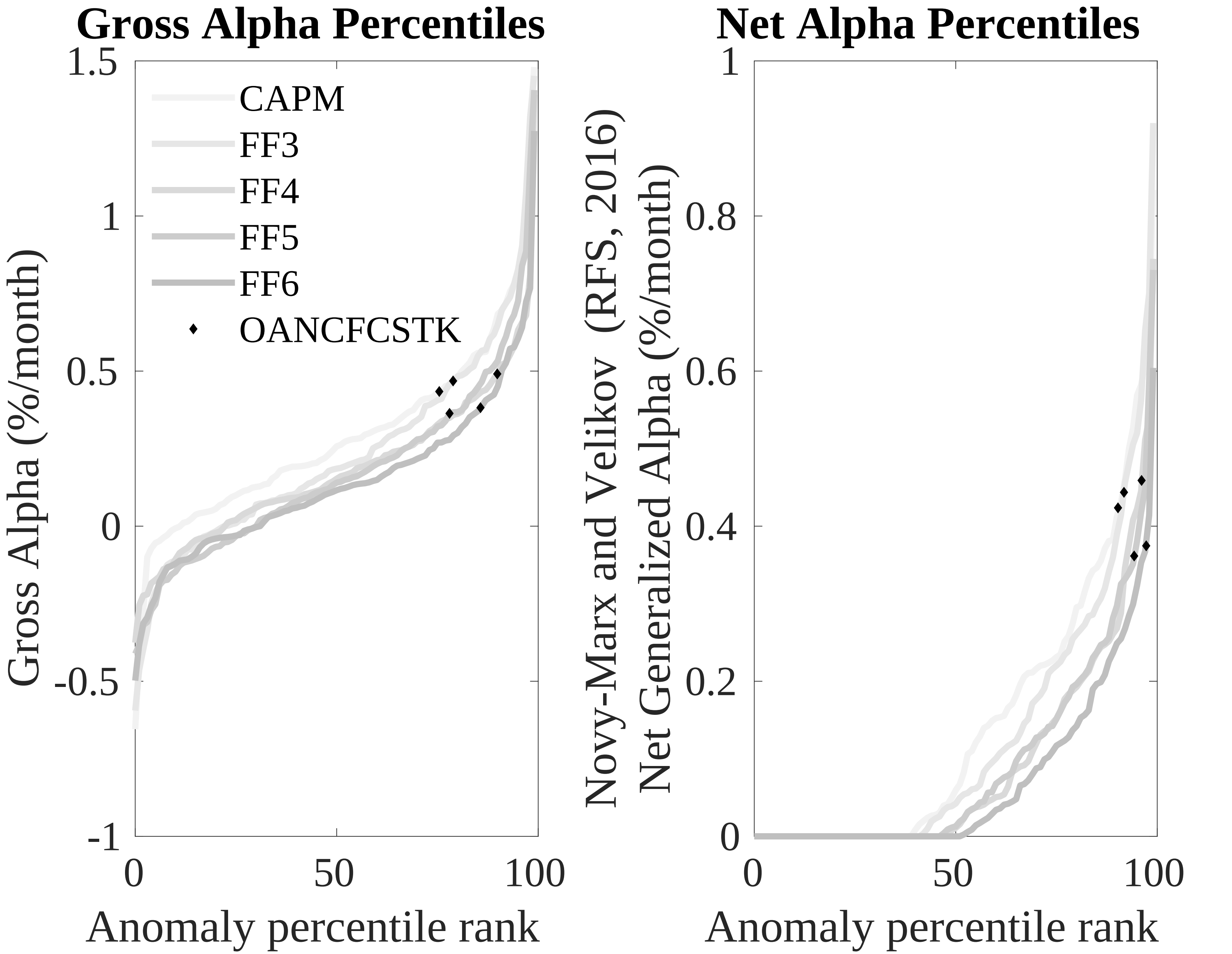
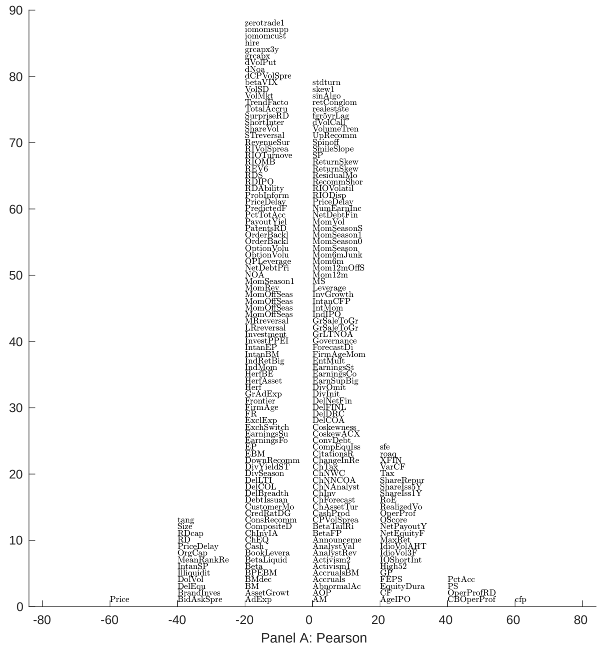


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SCP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
OANCFCSTK
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



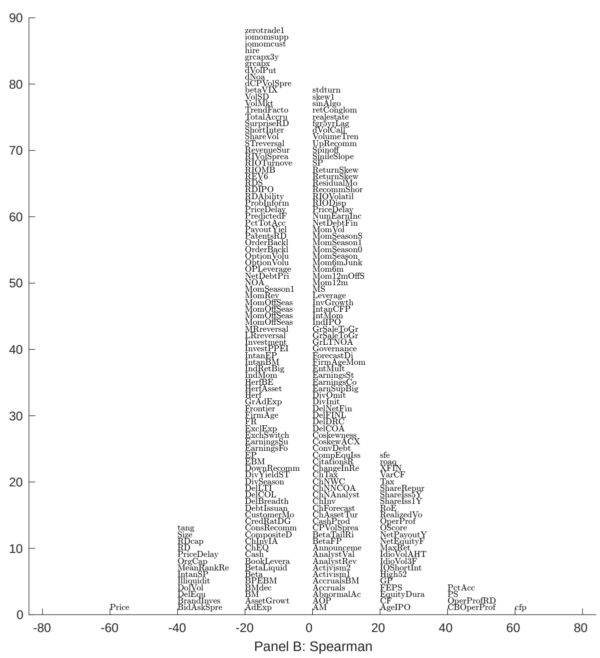


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with SCP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

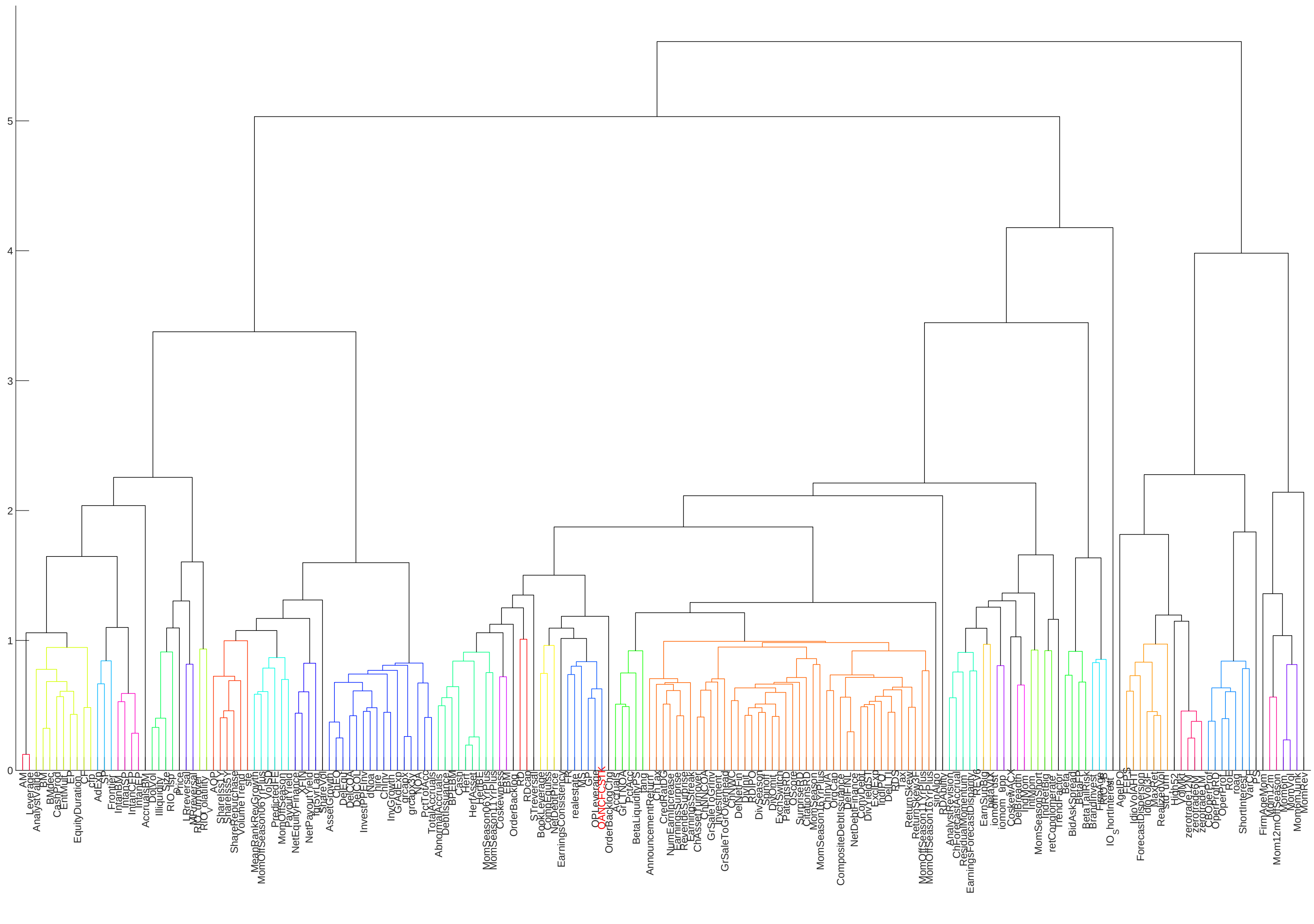


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

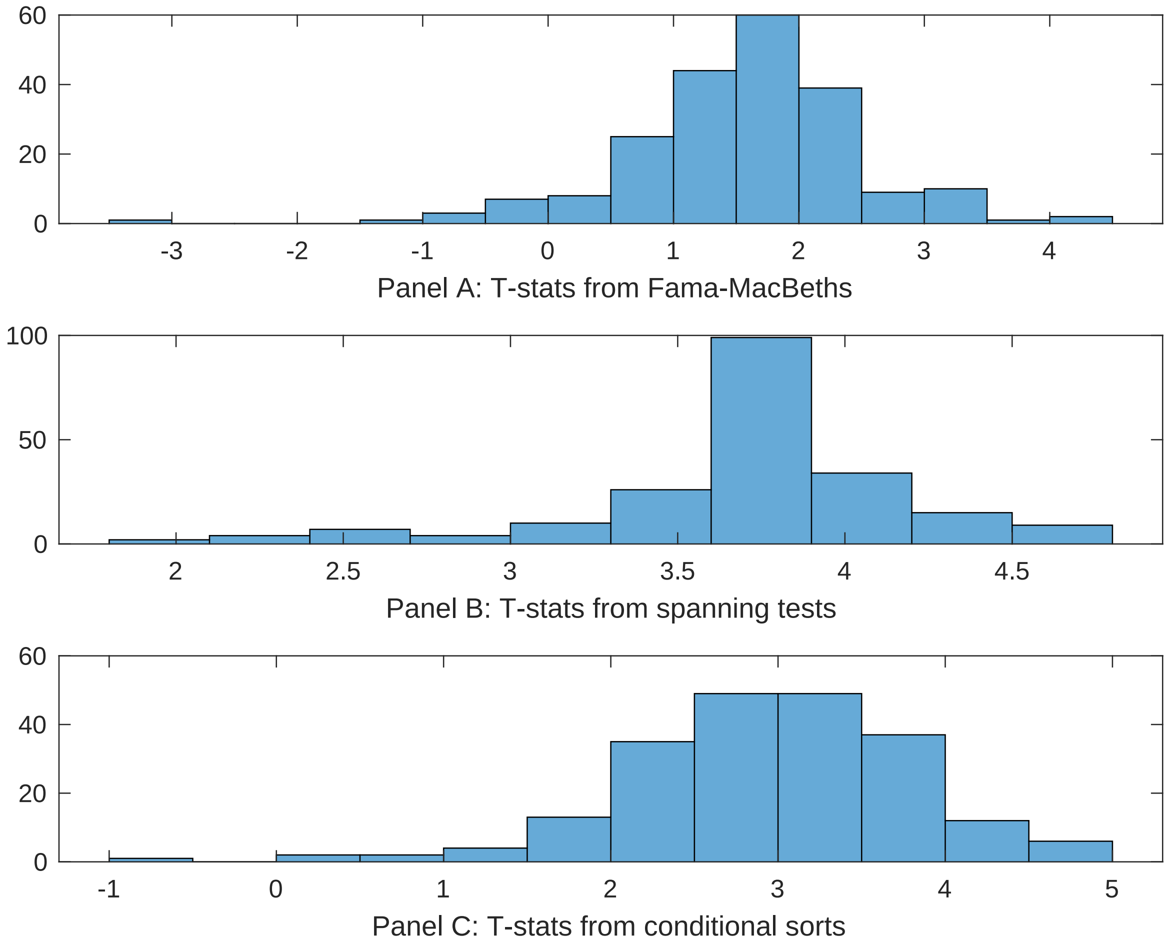


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of SCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SCP} SCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SCP. Stocks are finally grouped into five SCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SCP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SCP} SCP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.96	0.90	0.92	0.82	0.11	0.12	0.98
	[3.00]	[2.71]	[3.16]	[2.51]	[4.04]	[3.78]	[3.15]
SCP	0.56	0.20	0.30	0.18	0.82	0.30	0.12
	[0.54]	[1.76]	[2.59]	[1.44]	[0.92]	[2.69]	[1.27]
Anomaly 1	0.18						0.15
	[4.24]						[4.08]
Anomaly 2		0.17					-0.29
		[3.38]					[-0.61]
Anomaly 3			0.36				0.55
			[2.56]				[2.03]
Anomaly 4				0.72			0.43
				[3.79]			[1.87]
Anomaly 5					0.41		0.45
					[2.57]		[3.09]
Anomaly 6						0.74	0.76
						[2.48]	[2.79]
# months	427	427	427	432	427	427	427
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SCP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SCP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.28	0.29	0.38	0.31	0.35	0.38	0.23
	[2.71]	[2.78]	[3.58]	[2.98]	[3.31]	[3.52]	[2.21]
Anomaly 1	27.72						20.42
	[6.15]						[2.98]
Anomaly 2		19.61					-4.37
		[4.55]					[-0.70]
Anomaly 3			-8.91				-26.82
			[-1.08]				[-2.20]
Anomaly 4				25.75			16.20
				[6.06]			[3.18]
Anomaly 5					17.97		12.18
					[3.86]		[2.51]
Anomaly 6						3.26	32.66
						[0.44]	[3.01]
mkt	10.73	11.57	7.35	7.66	11.00	9.20	14.83
	[4.18]	[4.32]	[2.59]	[3.02]	[4.10]	[2.93]	[4.61]
smb	-14.15	-15.60	-11.14	-22.54	-17.24	-23.78	-15.84
	[-3.72]	[-4.01]	[-1.14]	[-6.18]	[-4.47]	[-3.11]	[-1.70]
hml	-18.14	-20.93	-27.62	-18.68	-23.06	-29.80	-16.59
	[-3.79]	[-4.30]	[-5.75]	[-3.91]	[-4.79]	[-5.95]	[-3.28]
rmw	17.89	17.36	29.86	18.25	15.54	31.86	-0.94
	[3.49]	[3.06]	[5.81]	[3.56]	[2.45]	[6.63]	[-0.14]
cma	-19.99	-15.73	-14.05	-10.32	-15.69	-14.80	-15.74
	[-3.06]	[-2.38]	[-2.08]	[-1.58]	[-2.36]	[-2.19]	[-2.39]
umd	-4.92	-5.17	-2.88	-3.64	-7.20	-2.13	-3.69
	[-2.15]	[-2.19]	[-1.22]	[-1.61]	[-2.77]	[-0.83]	[-1.34]
# months	428	428	428	432	428	428	428
$\bar{R}^2$(%)	39	37	34	40	36	34	42

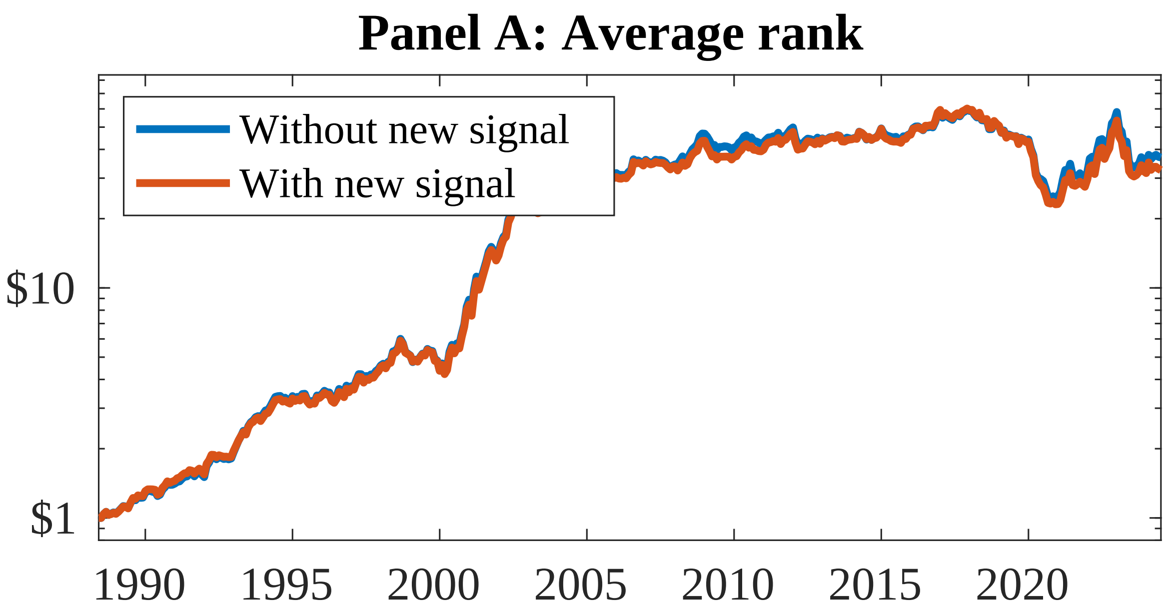


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as SCP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Ball, R., Gerakos, J., Linnainmaa, J. T., and Nikolaev, V. V. (2015). Deflating profitability. *Journal of Financial Economics*, 117(2):225–248.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2009). Driven to distraction: Extraneous events and underreaction to earnings news. *Journal of Finance*, 64(5):2289–2325.

Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.

Hou, K. and Moskowitz, T. J. (2005). Market frictions, price delay, and the cross-section of expected returns. *Review of Financial Studies*, 18(3):981–1020.

Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Peng, L. and Xiong, W. (2007). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80(3):563–602.

Richardson, S. A., Sloan, R. G., Soliman, M. T., and Tuna, I. (2005). Accrual reliability, earnings persistence and stock prices. *Journal of Accounting and Economics*, 39(3):437–485.

Sloan, R. G. (1996). Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3):289–315.

Zhang, X. F. (2006). Information uncertainty and stock returns. *Journal of Finance*, 61(1):105–137.